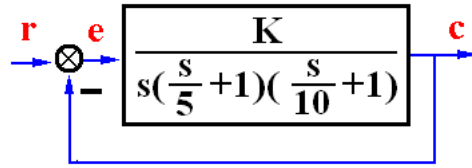


频率法串联迟后校正

例 1：系统如右，求 $G_c(s)$ 使：

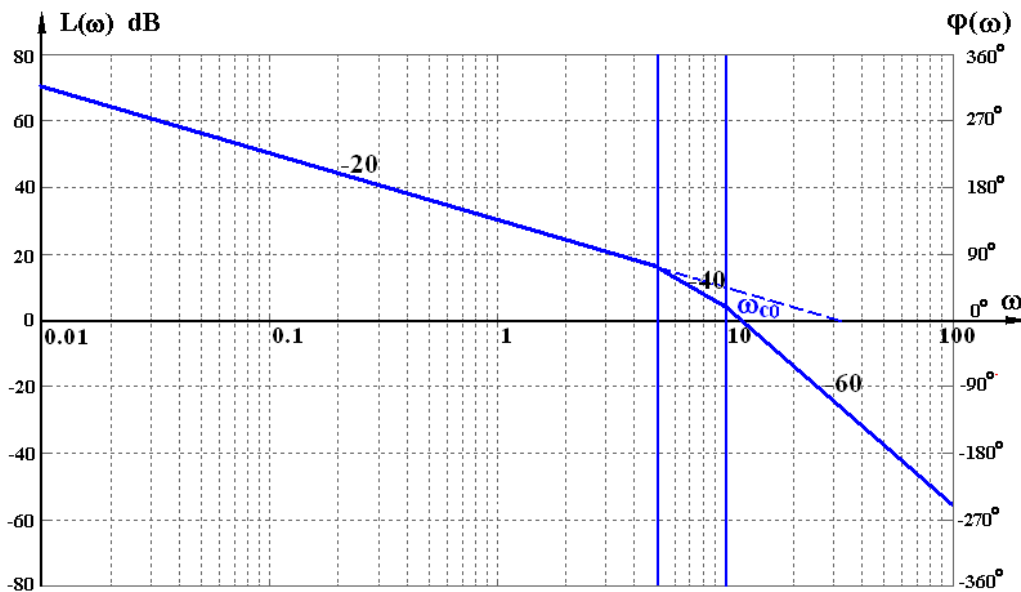
$$\begin{cases} K_v^* = 30 & 1/s \\ \gamma^* \geq 40^\circ \\ \omega_c^* \geq 2.3 & \text{rad/s} \end{cases}$$



解：① 由 e_{ss}^* 要求：取 $K = K_v^* = 30$

$$G_0(s) = \frac{K}{s(\frac{s}{5}+1)(\frac{s}{10}+1)} \quad \begin{cases} K = 30 \\ v = 1 \end{cases}$$

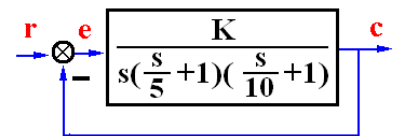
② 作 $L_0(\omega)$ 曲线：



$$|G(\omega_{c0})| = \frac{30}{\omega_{c0} \cdot \frac{\omega_{c0}}{10} \cdot \frac{\omega_{c0}}{5}} = 1 \Rightarrow \omega_{c0}^3 = 1500$$

$$\omega_{c0} = \sqrt[3]{1500} = 11.45 > \omega_c^* = 2.3$$

$$\begin{aligned} \gamma_0 &= 180^\circ - 90^\circ - \arctan(0.1\omega_{c0}) - \arctan(0.2\omega_{c0}) \\ &= 180^\circ - 90^\circ - 48.9^\circ - 66.4^\circ = -25.28^\circ < 40^\circ = \gamma^* \end{aligned}$$

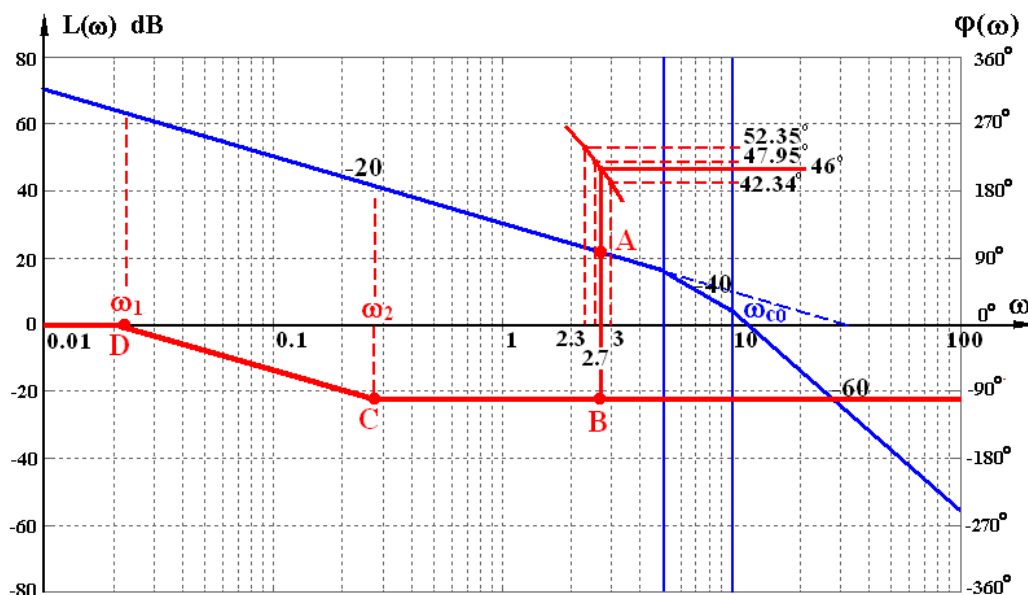


若采用超前校正： $\varphi_m = 40^\circ + 25.28^\circ + 10^\circ = 75.28^\circ > 60^\circ$

原系统特点： $\begin{cases} \omega_{c0} \text{ 有余} \\ \gamma_0 \text{ 不足} \end{cases} \longrightarrow \text{用迟后校正}$

③ 取 ω_{c1} 使： $\gamma_c(\omega_{c1}) = 180^\circ + \varphi_0(\omega_{c1}) = \gamma^* + 6^\circ = 46^\circ$

作 $\gamma_c(\omega)$ 曲线：分析可见： $\gamma_c(\omega) = 46^\circ \Rightarrow \omega < 5$



试探 ω_{c1} ： $\gamma_c = 180^\circ - 90^\circ - \arctan 0.1\omega - \arctan 0.2\omega$

$$\left. \begin{aligned} \gamma_c(2.3) &= 90^\circ - \arctan 0.23 - \arctan 0.46 = 52.345^\circ \\ \gamma_c(3) &= 90^\circ - \arctan 0.3 - \arctan 0.6 = 42.337^\circ \\ \gamma_c(2.6) &= 90^\circ - \arctan 0.26 - \arctan 0.52 = 47.951^\circ \end{aligned} \right\} \text{作 } \gamma_c(\omega)$$

$$\gamma_c = 46^\circ \Rightarrow \omega_{c1} = 2.7$$

可见 ω_c 的选择范围为： $2.3 \leq \omega_c \leq 2.7$

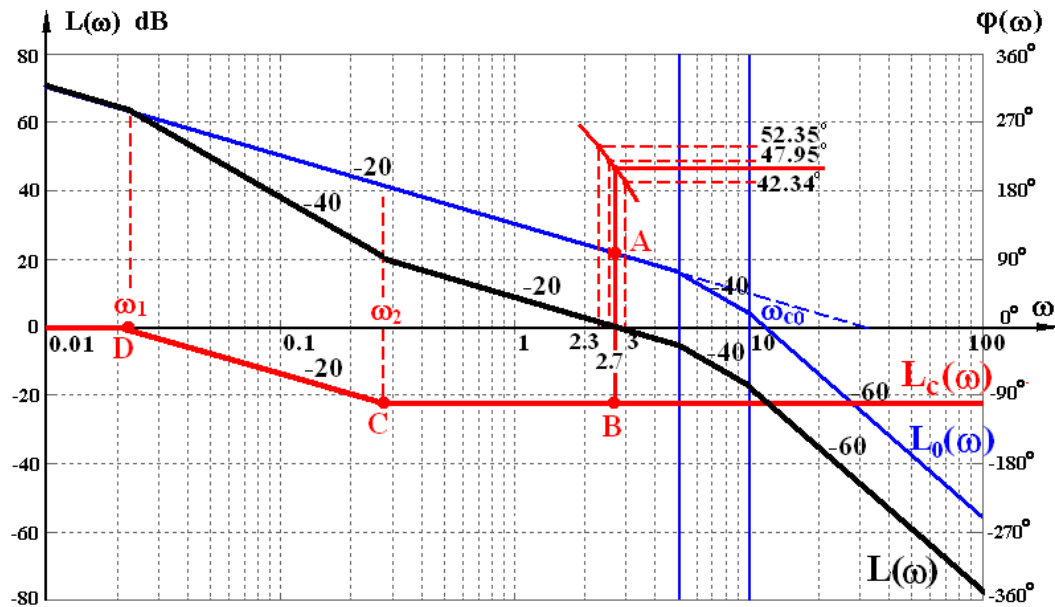
④ $\omega_c = 2.7 \Rightarrow A \rightarrow B \rightarrow C \rightarrow D$

$$\text{C 点: } \omega_2 = 0.1 \times 2.7 = 0.27$$

$$\text{D 点: } \frac{K}{\omega_c} = \frac{\omega_2}{\omega_1}, \quad \omega_1 = \frac{\omega_c \omega_2}{K} = \frac{2.7 \times 0.27}{30} = 0.0243$$

$$G_c(s) = \frac{\frac{s}{\omega_2} + 1}{\frac{s}{\omega_1} + 1} = \frac{\frac{1}{0.27}s + 1}{\frac{1}{0.0243}s + 1}$$

$$G(s) = G_c(s) \cdot G_0(s) = \frac{30}{s(\frac{s}{5} + 1)(\frac{s}{10} + 1)} \cdot \frac{\frac{s}{0.27} + 1}{\frac{s}{0.0243} + 1}$$



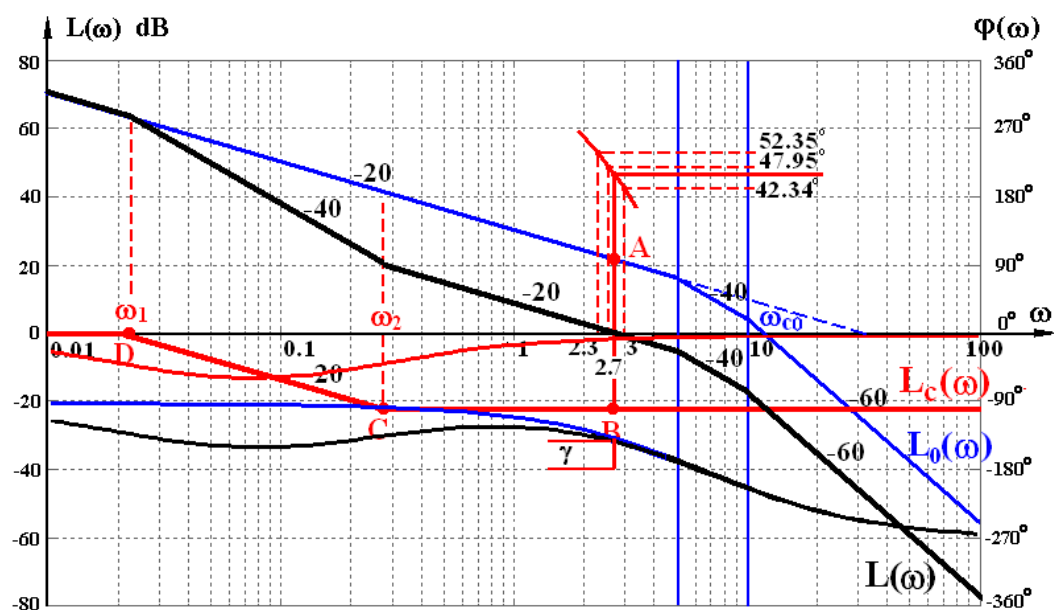
$$G(s) = G_c(s) \cdot G_0(s) = \frac{30}{s(\frac{s}{5} + 1)(\frac{s}{10} + 1)} \cdot \frac{\frac{s}{0.27} + 1}{\frac{s}{0.0243} + 1}$$

$$\omega_c = 2.7$$

$$\gamma = 180^\circ + \varphi(\omega_c)$$

$$= 180^\circ + \arctan \frac{2.7}{0.27} - 90^\circ - \arctan \frac{2.7}{5} - \arctan \frac{2.7}{10} - \arctan \frac{2.7}{0.0243}$$

$$= 180^\circ + 84.3^\circ - 90^\circ - 28.4^\circ - 15.1^\circ - 89.5^\circ = 41.3^\circ > \gamma^* = 40^\circ$$



⑤ 验算结果： $\left\{ \begin{array}{l} K = K_v = 30 \\ \gamma = 180^\circ + \angle G(\omega_c) = 41.3^\circ \geq 40^\circ \\ \omega_c = 2.7 > 2.3 \end{array} \right\}$ 满足要求。

注：
$$G_c(s) = \frac{\frac{s}{\omega_2} + 1}{\frac{s}{\omega_1} + 1} = \frac{\frac{s}{0.27} + 1}{\frac{s}{0.0243} + 1}$$

$$T_{max} = \frac{1}{0.0243} = 41.152 \text{ (sec)}$$

$$t_s = 3T_{max} = 123.5 \text{ (sec)} = 2.06 \text{ (min)}$$

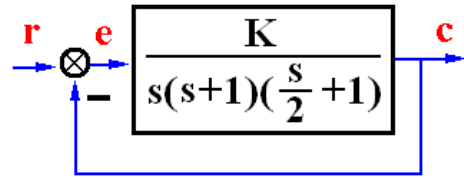
例 2 系统结构图如右图所示

要求：1) 静态速度误差 $e_{ss}^* \leq 0.1$

2) $\gamma^* \geq 40^\circ$

3) $20 \lg h^* \geq 10 \text{ dB}$

试确定校正装置传递函数 $G_c(s) = ?$



解： $G(s) = \frac{K}{s(s+1)(\frac{s}{2}+1)}$ $\begin{cases} \text{开环增益: } K \\ \nu = 1 \end{cases}$

① 依指标确定 K

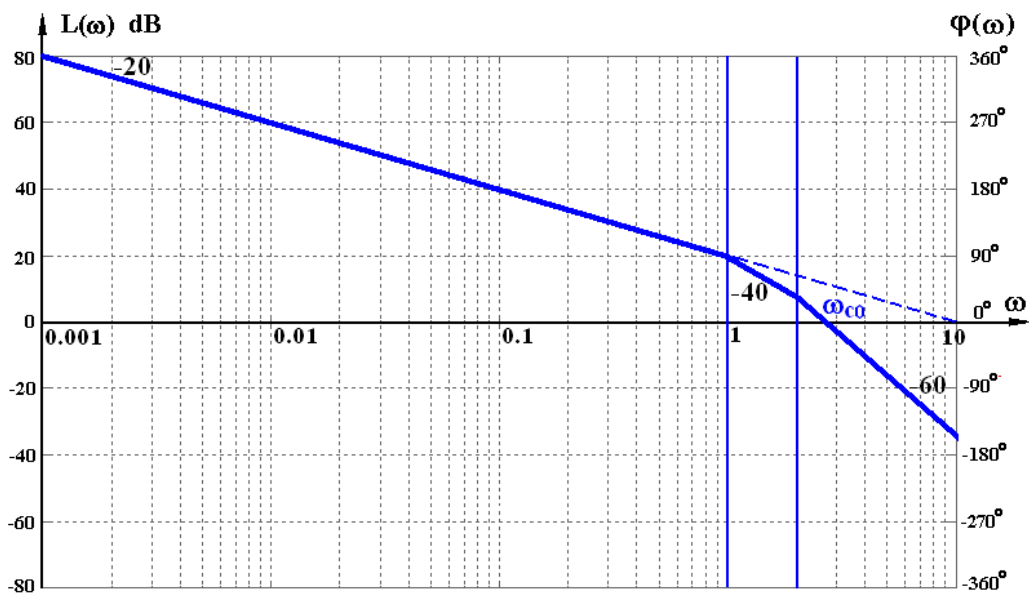
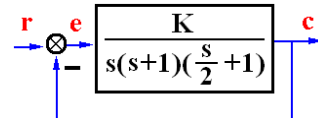
$$e_{ss}^{r=t} = \frac{1}{K} \leq 0.1 \Rightarrow K \geq 10, \text{ 取 } K = 10$$

② $L_0(\omega) \rightarrow \omega_{c0} \rightarrow \gamma_0$

$$|G_0(\omega_{c0})| = \frac{10}{\omega_{c0} \cdot \omega_{c0} \frac{\omega_{c0}}{2}} = \frac{20}{\omega_{c0}^3} = 1$$

$$\omega_{c0} = \sqrt[3]{20} = 2.7144$$

$$\begin{aligned} \gamma_0 &= 180^\circ - 90^\circ - \arctan \omega_{c0} - \arctan \frac{\omega_{c0}}{2} \\ &= 90^\circ - 68.8^\circ - 53.6^\circ = -33.4^\circ \end{aligned}$$



采用超前校正：

$$\varphi_m = \gamma^* - \gamma_0 + 10^\circ = 40^\circ + 33.4^\circ + 10^\circ = 83.4^\circ > 60^\circ$$

用一级超前校正不能解决问题，试用迟后校正（ ω_c^* 没有要求）。

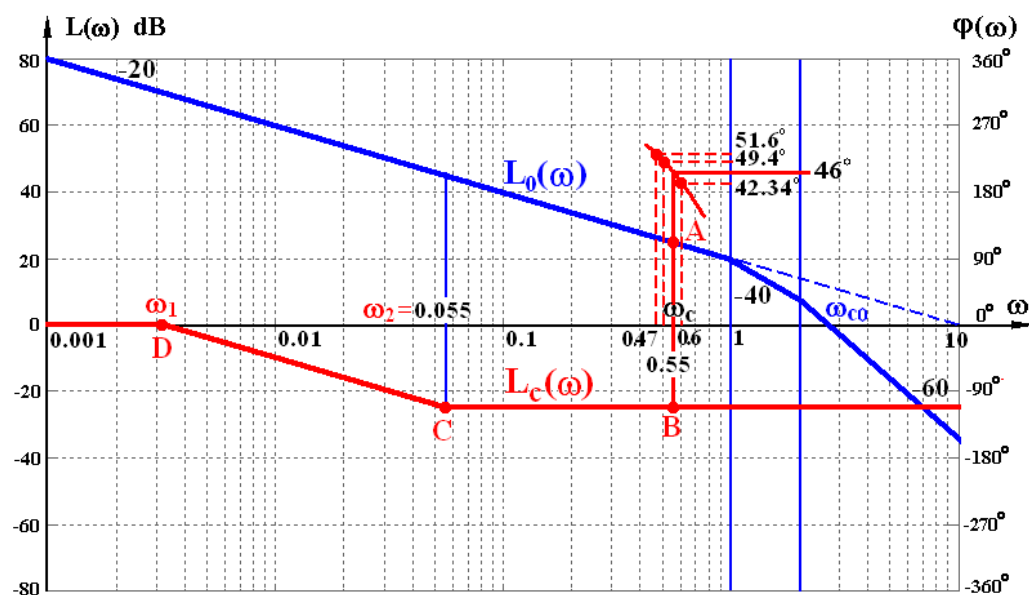
③ 选定 ω_{c1} ，依题，定 $\gamma_c(\omega_{c1}) = \gamma^* + 6^\circ = 40^\circ + 6^\circ = 46^\circ$

作 $\gamma_c(\omega)$ 曲线

ω	$\gamma_c(\omega) = 180^\circ + \angle G_0(j\omega)$
0.6	42.34°
0.5	49.4°
0.47	51.6°

作 $\gamma_c(\omega)$ 曲线， $\gamma_c = 46^\circ \Rightarrow \omega_{c1} = 0.55$

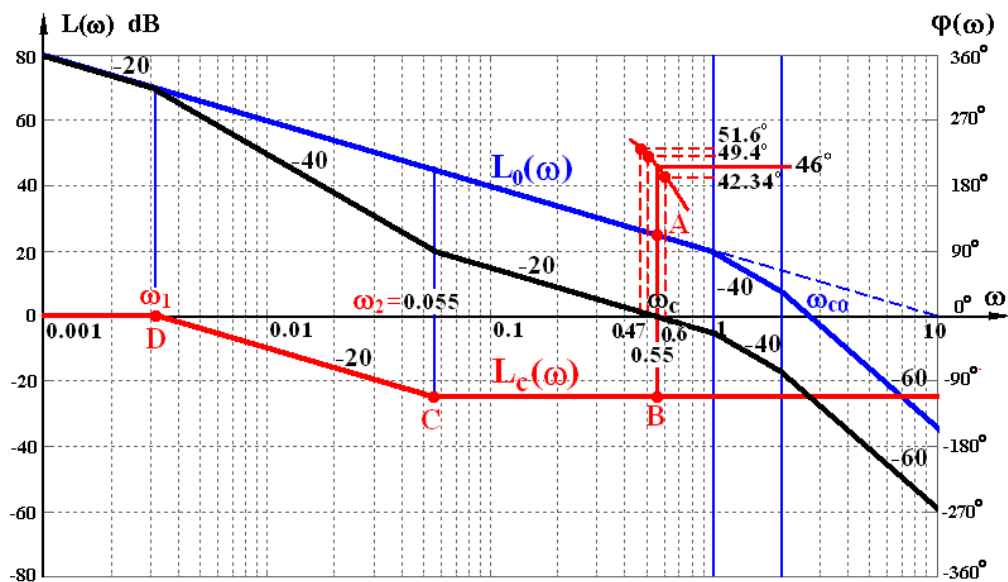
④ 取 $\omega_c = 0.55$ 作图设计：→A, B, C, D，确定 $G_c(s)$



C 点： $\omega_2 = 0.1\omega_c = 0.1 \times 0.55 = 0.055$

D 点： $\frac{10}{\omega_c} = \frac{\omega_2}{\omega_1}$ $\omega_1 = \frac{\omega_c \cdot \omega_2}{10} = \frac{0.55 \times 0.055}{10} = 0.003025$

$$G_c(s) = \frac{\frac{s}{\omega_2} + 1}{\frac{s}{\omega_1} s + 1} = \frac{\frac{s}{0.055} + 1}{\frac{s}{0.003025} + 1} \quad \begin{cases} b = \frac{0.055}{0.003025} = 18.18 \\ T = \frac{1}{\omega_1} = 330.6'' = 5.5(\text{min}) \end{cases}$$



⑤ 验算: $G(s) = G_c(s) \cdot G_0(s) = \frac{10(\frac{s}{0.055} + 1)}{s(s+1)(\frac{s}{2} + 1)(\frac{s}{0.003025} + 1)}$

$$\begin{aligned} \gamma &= 180^\circ + \arctan \frac{0.55}{0.055} - 90^\circ - \arctan 0.55 - \arctan \frac{0.55}{2} - \arctan \frac{0.55}{0.003024} \\ &= 180^\circ + 84.29^\circ - 90^\circ - 28.81^\circ - 15.38^\circ - 89.69^\circ = 40.42^\circ > 40^\circ \end{aligned}$$

检验 h 方法 I:

求 ω_g : 由 $L(\omega)$ 曲线, 应有 $\omega_g \approx \sqrt{2} = 1.414$ 。

试根: $\omega_g = 1.36$

$$|G(\omega_g)| = \frac{10 \sqrt{(\frac{1.36}{0.055})^2 + 1}}{1.36 \sqrt{(1.36)^2 + 1} \sqrt{(\frac{1.36}{2})^2 + 1} \sqrt{(\frac{1.36}{0.003025})^2 + 1}} = 0.1983$$

$$h = \frac{1}{0.1983} = 5.044$$

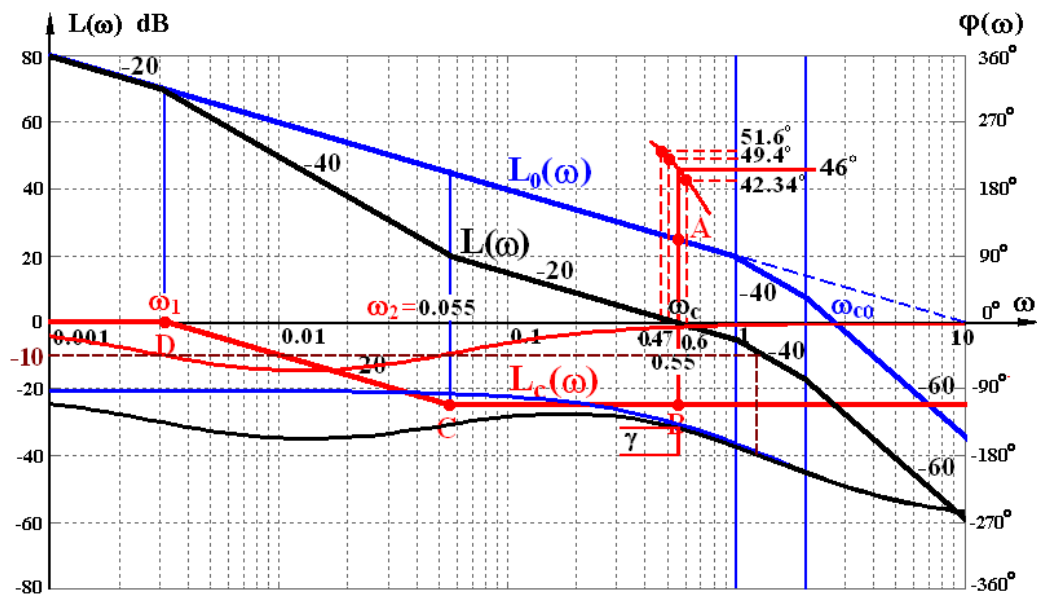
$$20 \lg h = 20 \lg 5.044 = 14.05 \text{ dB} > 10 \text{ dB}$$

检验 h 方法 II:

$$20 \lg K_h = 10 \text{ dB}, \quad K_h = 10^{\frac{10}{20}} = 3.16$$

$$|G(\omega_{g0})| = 3.16 \times \frac{10^{\frac{\omega_{g0}}{0.055}}}{\omega_{g0} \cdot \omega_{g0} \cdot \frac{\omega_{g0}}{0.003025}} = \frac{1.738}{\omega_{g0}^2} = 0$$

$$\begin{cases} \omega_{g0} = \sqrt{1.738} = 1.318 \\ \varphi(\omega_{g0}) = \varphi(1.38) = -178.5^\circ > -180^\circ \end{cases} \quad h > h^* = 10 \text{ dB}$$



迟后校正与超前校正效果的比较:

- 超前: $\begin{cases} \gamma_c \uparrow, \omega_c \uparrow \rightarrow \text{中频段特性得以改善, 动态性能改善} \\ L(\omega) \text{ 高频段} \uparrow \rightarrow \text{校正后系统抗高频干扰能力} \downarrow \end{cases}$
- 迟后: $\begin{cases} \gamma \uparrow, \omega_c \downarrow \rightarrow \text{牺牲快速性, 改善均匀性} \\ L(\omega) \text{ 高频段} \downarrow \rightarrow \text{校正后系统抗高频干扰能力} \uparrow \end{cases}$